

Overcurrent relays with inverse definite minimum time (IDMT) characteristics are widely used in power systems to protect electrical feeders and loads from overheating under abnormal

overcurrent relay is 51.

Although the proposed relay system is not thermal-based, it effectively emulates IDMT characteristics and is suitable for protecting induction motors, feeders, and other loads against prolonged overload conditions that could otherwise lead to winding failure or insulation damage. The implemented system, shown in Fig. 1, has been deployed on a domestic borewell motor to provide an alarm during overload conditions.

Fig. 2 shows the circuit diagram of the overcurrent/IDMT relay system. The system is built around an Arduino Nano, an ACS712 (30A) Hall-effect current sensor module, and a few supporting components. The load or feeder whose current is to be monitored is connected in series with the ACS712 sensor module.

The ACS712 operates on the Hall-effect principle and can measure both AC and DC currents. It can be connected directly in series with a 230V, 50Hz AC circuit, provided the current does not exceed the module's rated limit.

In the implemented system, a relay is used to monitor an induction motor whose starting current exceeds the 30A rating of the sensor module. Therefore, a 30A/5A, 5VA current transformer (CT) is used to scale down the current under all operating conditions.

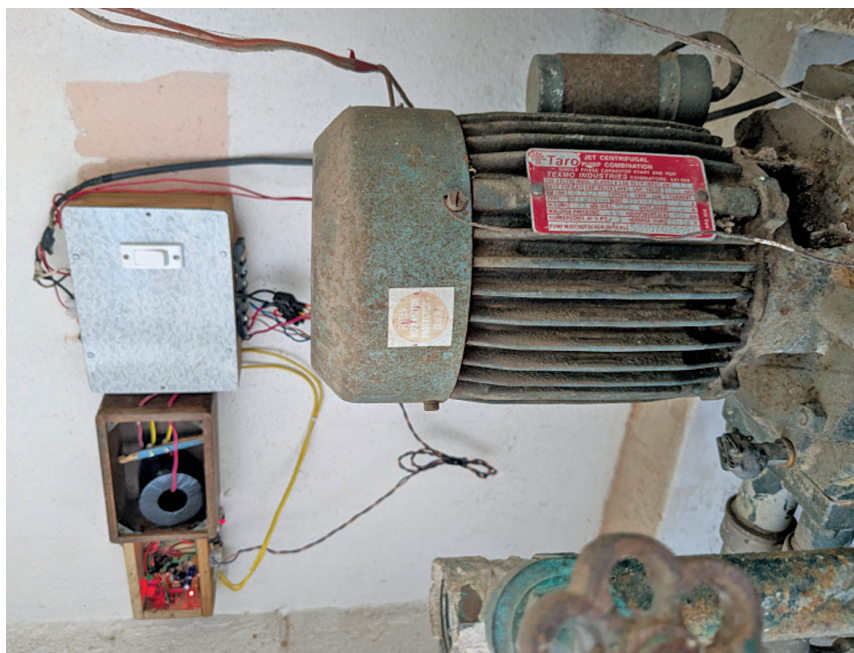
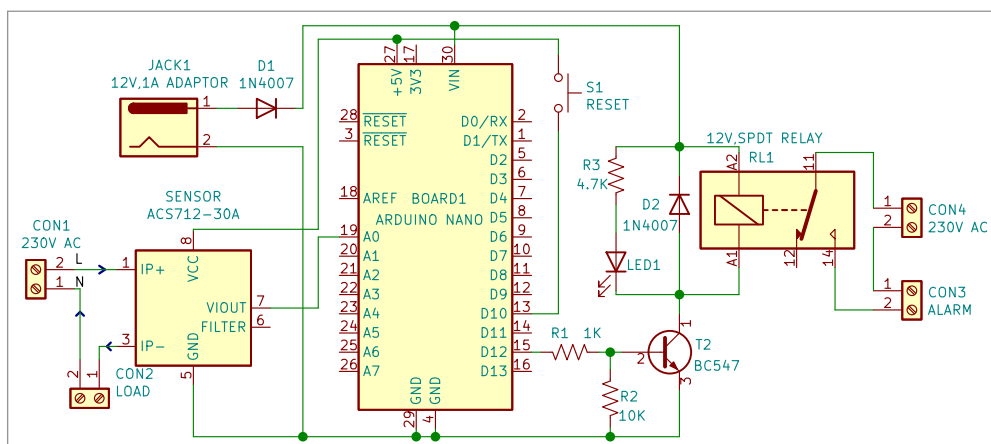


Fig. 1: Author's prototype



*Fig. 2: Circuit diagram*